```
def simulate(total_frames, window_size, lost_frame):
    base = 0
    next_seq = 0
    while base < total_frames:
        window_end = min(base + window_size - 1, total_frame - 1)
        print("\n----------")
        print(f"Sender window: [Frame {base} - Frame {window_end}]").

    while next_seq < base + window_size and next_seq < total_frames:
        print(f"Node A --> Sending Frame {next_seq}")
        if next_seq == lost_frame:
            print(f"(Frame {next_seq} LOST in transit!)")
            next_seq += 1.
        if lost_frame is not None and base <= lost_frame < next_seq:
            print(f"Node A --> Timeout! Resending window starting from frame {lost_frame}.")
            lost_frame = None
            continue.
        for f in range(base, next_seq):
            print(f"Node B -> Received Frame {f}, Sending ACK {f}")
```

Exp no: 07
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Flow control at Data Link Layer using Sliding Window Protocol.

Aim:-

To write a program to implement flow control at the Data Link layer using the sliding window protocol, and to simulate as ~~lost~~ flow of frames from one node to another.

Algorithm:-

1. Start the program
2. Read the total no. of frames to be sent (N), the sender's window size (W), and (optionally) a frame number to simulate as lost.
3. Initialize base = 0 and next seq = 0.
4. Repeat while base < N:
   a. Determine the current sender window as range
   b. Send all the frames with sequence no from nextseq up to (base + W - 1),
   c. If the lost frame lies with the frames.
      (i) receiver discards the lost frame.
      (ii) sender timer out & retransmits all frames
   d. Else
      (i) The receiver accepts each frame in order. and sends a cumulative acknowledgment for it.
      (ii) On receiving the ACK for the last frame sent.

```
print("\n----------")
print(f"All {total_frames} frames successfully
  transmitted from Node A to Node B.")

def main():
  print("=== SLIDING WINDOW PROTOCOL -
  FLOW CONTROL SIMULATION (GO-back-N)
  ===\n")
  total_frames = int(input("Enter total number
  of frames to send : "))
  window_size = int(input("Enter sender's
  window size : "))
  if lost_frame == -1:
    lost_frame = None
  simulate(total_frame, window_size,
      lost_frame)

if __name__ == "__main__":
  main()
```

OUTPUT

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send    : 8
Enter sender's window size              : 4
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss)          : 5

---------------------------------------
Sender Window : [Frame 0 - Frame 3]
Node A -->Sending Frame 0
Node A -->Sending Frame 1
Node A -->Sending Frame 2
Node A -->Sending Frame 3
Node B -->Received Frame 0, sending ACK 0
Node B -->Received Frame 1, sending ACK 1
Node B -->Received Frame 2, sending ACK 2
Node B -->Received Frame 3, sending ACK 3
Node A -->ACK 3 received. Sliding window forward.

---------------------------------------
Sender Window : [Frame 4 - Frame 7]
Node A -->Sending Frame 4
Node A -->Sending Frame 5
        (Frame 5 LOST in transit!)
Node A -->Sending Frame 6
Node A -->Sending Frame 7
Node B -->Frame 5 missing. Discarding out-of-order frames after it.
Node A -->Timeout! Resending window starting from Frame 5.

---------------------------------------
Sender Window : [Frame 4 - Frame 7]
Node A -->Sending Frame 5
Node A -->Sending Frame 6
Node A -->Sending Frame 7
Node B -->Received Frame 4, sending ACK 4
Node B -->Received Frame 5, sending ACK 5
Node B -->Received Frame 6, sending ACK 6

Node B -->Received Frame 7, sending ACK 7
Node A -->ACK 7 received. Sliding window forward.

---------------------------------------
All 8 frames successfully transmitted from Node A to Node B.
```

Result:-

The program to implement flow control at the Data Link layer using the sliding window protocol was successfully written in python executed and verified.